

Retail Market Basket Intelligence

Phase 2: Data Preprocessing & Transformation Pipeline

Prepared By:

Karthik Yelugam | Data Analyst

Tools Deployed:

- Python (Pandas, NumPy, SQLAlchemy)
- MySQL Workbench

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1. EXECUTIVE SUMMARY

This document presents the Phase 2 engineering documentation for the Retail Market Basket Intelligence project. The primary objective of this phase is to transform the raw retail transaction dataset into a clean, consistent, and mathematically structured dataset required for Market Basket Analysis utilizing the Apriori Algorithm.

Raw retail transaction datasets inherently contain data quality issues such as missing values, duplicate rows, incorrect numerical values, cancelled transactions, and inconsistent text formatting. These anomalies can severely skew the accuracy of analytical results if not systematically mitigated. Therefore, a rigorous data cleaning and transformation pipeline was implemented to ensure the dataset's absolute reliability.

The workflow of this phase commenced with dataset validation and quality assessment to isolate structural problems within the raw data. Following identification, sequential data cleansing operations were executed. These included handling missing values, purging duplicate records, filtering cancelled orders, correcting negative quantities and price anomalies, and globally standardizing product and country nomenclature. The temporal column was also parsed into a structured datetime format to support advanced time-based analytics.

Post-cleansing, the data was enriched through feature engineering. Newly derived metrics-such as *Revenue*, *Transaction Size*, and *Product Frequency*-were generated to map transaction behavior and product demand accurately.

Finally, the dataset underwent structural transformation into a sparse boolean matrix. Transactions were aggregated by invoice number, converted into a binary basket matrix, and algorithmically filtered to remove ultra-rare products, thereby optimizing computational efficiency and rule quality.

The complete pipeline was orchestrated using MySQL Workbench for database validation, Python (Pandas, NumPy) for programmatic transformation, and Microsoft Excel for output verification. This report serves as the comprehensive log for dataset validation, data cleaning, feature engineering, and matrix preparation prior to algorithmic deployment in Phase 3.

2. DATASET OVERVIEW

The dataset consists of retail transaction records extracted from the secure relational database table.

Dataset Size

Attribute	Value
Total Rows	522,064
Total Columns	7

Dataset Schema (Columns)

Column Name	Description
BillNo	Unique invoice number representing a transaction
Itemname	The specific product purchased
Quantity	Number of units purchased per line item
Present_Date	Transaction timestamp
Price	Unit price of the product
CustomerID	Unique identifier of the purchasing customer
Country	Geographic location where the purchase occurred

3. INITIAL DATASET PROFILE

Before the execution of the cleaning pipeline, the raw dataset contained multiple critical data quality issues that required programmatic resolution.

Summary of Initial Anomalies

Issue Type	Result Found
Total Baseline Rows	522,064
Missing Itemname	1,455 rows
Missing CustomerID	134,041 rows
Duplicate Rows	5,286 rows
Negative Quantities	1,336 rows
Zero or Negative Prices	2,513 rows

Conclusion: These issues would significantly degrade the accuracy of the Apriori association algorithms, mandating a systematic cleansing protocol.

4. DATA CLEANING PIPELINE

Each cleaning operation was performed sequentially in Python, validated against parallel SQL queries.

Dataset Size Changes Summary

Cleaning Step	Rows Before	Rows After
Initial Raw Dataset	522,064	522,064
Remove Missing Product Names	522,064	520,609
Handle Missing CustomerID	520,609	520,609
Remove Duplicate Rows	520,609	515,323
Handle Negative Quantities	515,323	514,850
Remove Quantity Outliers	514,850	460,723
Fix Price Anomalies	460,723	430,127
Clean Product Names	430,127	430,127
Standardize Country Names	430,127	430,127
Convert Date to Datetime	430,127	430,127

4.1 Remove Missing Product Names

- **Problem:** Some transactions lacked product identifiers (Itemname).
- **Validation Result:** 1,455 rows missing Itemname.
- **Action:** Rows missing critical item identifiers were purged, as they cannot be evaluated in Market Basket Analysis.
 - *Before:* 522,064 rows → *After:* 520,609 rows

4.2 Handle Missing CustomerID

- **Problem:** Substantial volume of transactions lacked customer identifiers.
- **Validation Result:** 134,041 rows missing CustomerID.
- **Action:** To prevent massive data loss, missing values were imputed using a placeholder value: "Guest".
 - *Before:* 520,609 rows → *After:* 520,609 rows (*No rows removed*)

4.3 Remove Duplicate Rows

- **Problem:** Identical duplicate transaction rows were detected.
- **Validation Result:** 5,286 duplicate rows. These distort product frequencies and artificially inflate Apriori support calculations.
- **Action:** Exact duplicates were dropped utilizing the Pandas `drop_duplicates()` function.
 - *Before:* 520,609 rows → *After:* 515,323 rows

4.4 Handle Negative Quantities

- **Problem:** Rows displaying negative purchase quantities.
- **Validation Result:** 1,336 rows. These represent product returns or cancellations.
- **Action:** Rows with quantities ≤ 0 were removed to isolate true purchase intent.
 - *Before:* 515,323 rows → *After:* 514,850 rows

4.5 Remove Quantity Outliers

- **Problem:** Extreme volumetric outliers were detected during statistical profiling. These distort averages and skew the transaction analysis toward wholesale behavior.
- **Action:** Statistical filtering via the Interquartile Range (IQR) technique was applied to isolate standard consumer variance.
 - *Before:* 514,850 rows → *After:* 460,723 rows

4.6 Fix Price Anomalies

- **Problem:** Invalid pricing structures.
- **Validation Result:** 2,513 rows displaying zero or negative unit prices.
- **Action:** Rows containing non-commercial pricing (refunds, zero-dollar freebies) were purged.
 - *Before:* 460,723 rows → *After:* 430,127 rows

4.7 Clean Product Names

- **Problem:** Product names suffered from formatting inconsistencies (leading/trailing spaces, varied capitalization).
- **Action:** Applied string cleaning operations: trimmed whitespaces, converted text strictly to UPPERCASE, and stripped anomalous special characters.
 - *Before:* 430,127 rows → *After:* 430,127 rows (*No rows removed*)

4.8 Standardize Country Names

- **Problem:** Geographic country names possessed formatting variations.
- **Action:** Standardized values by trimming whitespaces, applying consistent Title Case, and correcting spelling variations.
 - *Before: 430,127 rows → After: 430,127 rows (No rows removed)*

4.9 Convert Date Column to Datetime

- **Problem:** Present_Date was loaded as a raw text string.
- **Action:** Converted the string array into an optimized Pandas datetime format to enable temporal grouping.
 - *Before: 430,127 rows → After: 430,127 rows (No rows removed)*

5. FEATURE ENGINEERING

New analytical features were synthesized to enrich downstream dashboarding and model interpretability.

5.1 Revenue Column

- **Formula:** $\text{Revenue} = \text{Quantity} \times \text{Price}$
- **Purpose:** To analyze true transaction value and measure individual product revenue contributions.
- **Sample Calculation:**

Quantity	Price	Revenue
6	2.55	15.30

5.2 Transaction Size

- **Definition:** Represents the total number of distinct products purchased within a single transaction (basket size).
- **Purpose:** To analyze and group basket size distributions.
- **Sample Result:**

BillNo	Products	TransactionSize
536365	WHITE HANGING HEART TLIGHT HOLDER	7
536365	WHITE METAL LANTERN	7

5.3 Product Frequency

- **Definition:** Calculates how often each unique product line appears across the entire dataset.
- **Purpose:** To programmatically identify anchor products and detect ultra-rare products for filtering.
- **Sample Result:**

Product	Frequency
WHITE HANGING HEART TLIGHT HOLDER	1886
WHITE METAL LANTERN	287

6. PREPARATION FOR MARKET BASKET ANALYSIS

The structured dataset required pivoting into a specialized binary matrix format to be ingestible by the Apriori algorithm.

6.1 Transaction Basket Creation

Each transaction was aggregated by BillNo to formulate discrete itemsets.

- **Sample Result:**

BillNo	Itemname	Item_Count
536365	[WHITE HANGING HEART TLIGHT HOLDER, WHITE METAL LANTERN, ...]	7

6.2 Basket Matrix Creation

The grouped baskets were pivoted into a sparse binary Boolean matrix.

- **Sample Matrix Output:**

BillNo	10 COLOUR SPACEBOY PEN	12 COLOURED PARTY BALLOONS	12 DAISY PEGS IN WOOD BOX
536365	0	0	0
536366	0	0	1

- **Basket Matrix Summary Table:**

Description	Explanation
Rows	Represent individual transactions (baskets)
Columns	Represent individual unique products
Values	1 → Product present in transaction 0 → Product absent from transaction

6.3 Rare Product Filtering

- **Initial State:** The pivoted matrix contained **3,758** unique products.
- **Action:** Inclusion of ultra-rare items creates severe computational noise and yields statistically insignificant association rules. An algorithmic filter dropped products appearing fewer than 100 times.
- **Final State:** Feature space optimally reduced to **1,295** high-frequency products.

7. FINAL DATASET VALIDATION

A final suite of programmed validation checks was executed to guarantee data integrity prior to export.

Validation Check	Result
Missing Values	0
Duplicate Rows	0
Negative Quantities	0
Invalid Prices	0
Correct Data Types	Verified
Basket Matrix Structure	Verified

8. FINAL OUTPUT DATASETS

The pipeline bifurcates into two distinct production-ready datasets exported to local storage.

8.1 The Master Cleaned Dataset (cleaned_dataset.csv)

- **Structure:** BillNo, Itemname, Quantity, Present_Date, Price, CustomerID, Country, Revenue, TransactionSize, ProductFrequency.
- **Purpose:** Preserved for BI visualization, executive dashboards, and extended statistical validation.

8.2 The Apriori Boolean Matrix (apriori_dataset.csv)

- **Structure:** BillNo, Product_1, Product_2, Product_3...
- **Values:** Strictly Boolean integers (1 or 0).
- **Purpose:** The direct, optimized input dataset for the Apriori Machine Learning Algorithm.

9. CONCLUSION

Phase 2 successfully executed an enterprise-grade data engineering and transformation pipeline. The raw transactional data has been fully cleaned, standardized, mathematically validated, engineered with new business features, transformed into a binary transaction matrix, and securely exported.

The datasets are structurally flawless and computationally optimized. The pipeline is now fully prepared to advance into **Phase 3: Market Basket Analysis**, where the Apriori Algorithm will ingest the binary matrix to discover high-confidence itemsets and actionable association rules.